BIOLOGICAL SCIENCE
(Human Anatomy)

I. NERVOUS SYSTEM

- **Neurons –** basic cellular unit of the nervous system
- **Sensory neuron –** transmits impulses from the receptors to the central nervous system; main neuron system of the sensory organs
- **Interneuron –** located mainly in the central nervous system to interpret and relay nerve impulses between sensory motor neurons
- **Motor neuron –** transmit impulses form the central nervous system to effectors (glands and muscles) causing them to take action and respond
- **Nerves –** of neurons specialized for long distance and high speed impulse transmission
- **Myelin sheath –** covers neurons for insulation and protection
- **Neurotransmitters –** chemical messengers that produce certain effect on organs to maintain function.

Brain Large mass of neurons located in the cranial cavity; controls and coordinates all human activity	**Cerebrum**	of voluntary activity memory and thinking center
	Cerebellum	coordinates muscle movement and maintains balance
	Medulla	controls involuntary activities such as breathing, heartbeat, blood pressure and peristalsis
Spinal Cord	Protected by the vertebrate; center of reflex actions	

II. ENDOCRINE SYSTEM

- **Endocrine glands –** ductless gland located in various parts of the body secreting hormones
- **Hormones –** chemical mediators which target body organs to elicit certain response; transported from the endocrine glands to the target organ through the blood.
- **Hypothalamus –** links the nervous system to the endocrine system

Pituitary gland	**Growth stimulating hormone**	Elongation of long bones and muscle growth
	Thyroid stimulating hormone	Stimulates the thyroid gland to produce thyroxine
	Follicle stimulating hormone	Stimulates activity in the ovaries and testes
Thyroid gland	Produces THYROXINE which regulates rate of body metabolism	
Parathyroid gland	Secretes PARATHORMONE for returning calcium to blood circulation from the bones	

Adrenal glands	**Adrenal cortex**	Secretes CORTISOL for conversion of fat and proteins into glucose and ALDOSTERONE for reabsorption of sodium and chloride into the bloodstream.
	Adrenal medulla	Secretes ADRENALINE to increase the blood sugar level and accelerates the heart and breathing rate.

Testes	Male development of male sex characteristics.
Ovaries	Female sex gland which secretes testosterone to influence organ which secretes various hormones such as the estrogen which influences development of female sex characteristics.

III. CIRCULATORY SYSTEM

- **Red blood cells –** have hemoglobin; carry oxygen to various parts of the body
- **White blood cells –** fight infection in the body
- **Platelets –** needed for cessation of bleeding through blood clot formation
- **Arteries –** thick walled, muscular blood vessels which transport blood away from the heart to all parts of the body
- **Capillaries –** found at the end of small arteries, and at the beginning of small veins; exchanges dissolved materials by diffusion between the blood and fluid surrounding body cells
- **Veins –** thin walled blood vessels possessing valves which prevent back flow of blood; returns blood to the heart.

Heart Four chambered double pump composed of TWO atrium and TWO ventricles	**Right atrium**	Receives DEOXYGENATED blood from the body through the SUPERIOR VENA CAVA
	Left atrium	Receives OXYGENATED blood from the lungs through the PULMONARY VEIN
	Right ventricle	Pumps DEOXYGENATED blood to the lungs through the PULMONARY ARTERY
	Left ventricle	Pumps OXYGENATED blood to the rest of the body through the AORTA

Pulmonary circulation	Systemic circulation
Circulation to and from the lungs	Circulation to and from the rest of the body

IV. RESPIRATORY SYSTEM: involves cellular respiration and gas exchange

- **Cellular respiration –** process wherein oxygen is acquired by cells and processed to produce energy; end products are water and carbon dioxide
- **Gas exchange –** transportation of gases between the external environment and the internal membranes of the lungs
- **Nasal cavity –** lined with ciliated mucous membrane which filters, warms and moistens the air; opening is called nostrils.
- **Pharynx –** connects the nasal cavity to the air cavity; air travels here and passes the epiglottis, a flap of tissue which prevents other materials aside from air to enter the trachea
- **Trachea –** tube that sends air between the pharynx and the bronchi; cartilage rings prevent the trachea from collapsing
- **Bronchi –** lined with mucous membranes and ringed with cartilage leading to the bronchioles
- **Bronchioles –** lined with mucous membranes but lack cartilage which finally leads to alveoli
- **Alveoli –** functional unit where gas exchange occurs; surrounded by capillaries

Inhalation	Exhalation
• Diaphragm contracts and moves down to move air inside the lungs • As inhalation start air pressure inside the lungs is lower than environment	• Diaphragm relaxes and moves upward to move air outside the lungs • As exhalation starts air pressure inside the lungs is higher than the environment

V. DIGESTIVE SYSTEM: consists of a continuous "one-way" gastrointestinal (GI) tract and accessory organs

- **Peristalsis -** rhythmic muscular contractions to move food along the GI tracts
- **Ingestion –** digestion of food through the mouth which contains the teeth, tongue and salivary glands; serves to increase surface area of food for easier digestion
- **Movement of food –** mouth → esophagus → stomach → small intestines → large intestines → anus
- **Amylase –** digests starch into simple sugar
- **Stomach –** temporary storage area of food; where protein digestion begins through enzyme protease
- **Small intestine –** major portion of the food is digested; absorption of nutrients to the blood stream occurs through its VILLI structures
- **Gallbladder –** stores bile that helps in fat digestion
- **Pancreas –** produces protease, lipase and amylase which aid protein, lipid and starch digestion
- **Large intestine –** where water is mainly reabsorbed

VI. EXCRETORY SYSTEM

- **Liver –** breaks down red blood cells and recycle useable materials inside the body

- **Sweat glands –** through these structures of the skin water, salts and urea diffuses from the blood to the external surface of the body as perspiration
- **Kidneys –** excretion of urea; controls concentration of body fluids in the body
- **Nephrons –** functional unit for fluid filtration and reabsorption
- **Ureters –** two tubes which connects the kidneys to the urinary bladder
- **Urinary bladder –** stores urine until eliminated through the urethra
- **Urethra –** small tube where urine is finally excreted; contents of the urinary bladder empties to the urethra

VII. MUSCUKO-SKELETAL SYSTEM: usually operates in pair which pulls on the bones on either side of a joint.

Visceral muscles	Involuntary in action and smooth in appearance
Cardiac muscles	Involuntary in action and striated
Skeletal muscles	Striated and voluntary

- **Joint –** point of motion between two bones
- **Tendons –** attach muscles to bones
- **Ligaments –** connect ends of bones at movable joints

Function of bones:

1. Support
2. Protection
3. Anchorage sites for muscle action
4. Leverage for body motion
5. Production of blood cells (bone marrow)